EXP 4(b)	Advanced Data Modelling and Form Design

Aim:

To implement advanced data modelling in the Hotel Room Booking application using a Lookup relationship, Auto Number field, auto-calculated fields and Deluge validations.

Theory:

Advanced data modelling links related forms and reduces duplicate data entry. In this application the Room field of the Room Booking Form is a Lookup to the Rooms form (one room - many bookings). Booking ID is an Auto Number field starting at 1001.

When a room is selected, a workflow fetches the Room Type and Rate Per Night from the Rooms form, and Nights and Total Amount are calculated automatically (Nights = Check-Out - Check-In; Total = Nights x Rate).

Procedure:

1. Open the Room Booking Form in the form builder.
2. Configure the Room field as a Lookup to the Rooms form displaying Room Number.
3. Configure Booking ID as an Auto Number field with start index 1001.
4. Add Rate Per Night, Nights and Total Amount fields in the Billing Details section.
5. Create the user-input workflow on Room to fill Room Type and Rate and calculate the bill.
6. Open the live form, select a room and dates and verify the auto-filled values.

Deluge - On user input of Room (Fill Room Details):

```
// Fill room type and rate from the selected room, then recalculate the bill
if(input.Room != null)
{
    rm = Rooms[ID == input.Room];
    input.Room_Type = rm.Room_Type;
    input.Rate_Per_Night = rm.Rate_Per_Night;
    if(input.Check_In_Date != null && input.Check_Out_Date != null)
    {
        input.Nights = input.Check_In_Date.daysBetween(input.Check_Out_Date);
        input.Total_amount = input.Nights * rm.Rate_Per_Night;
    }
}
```

Output:

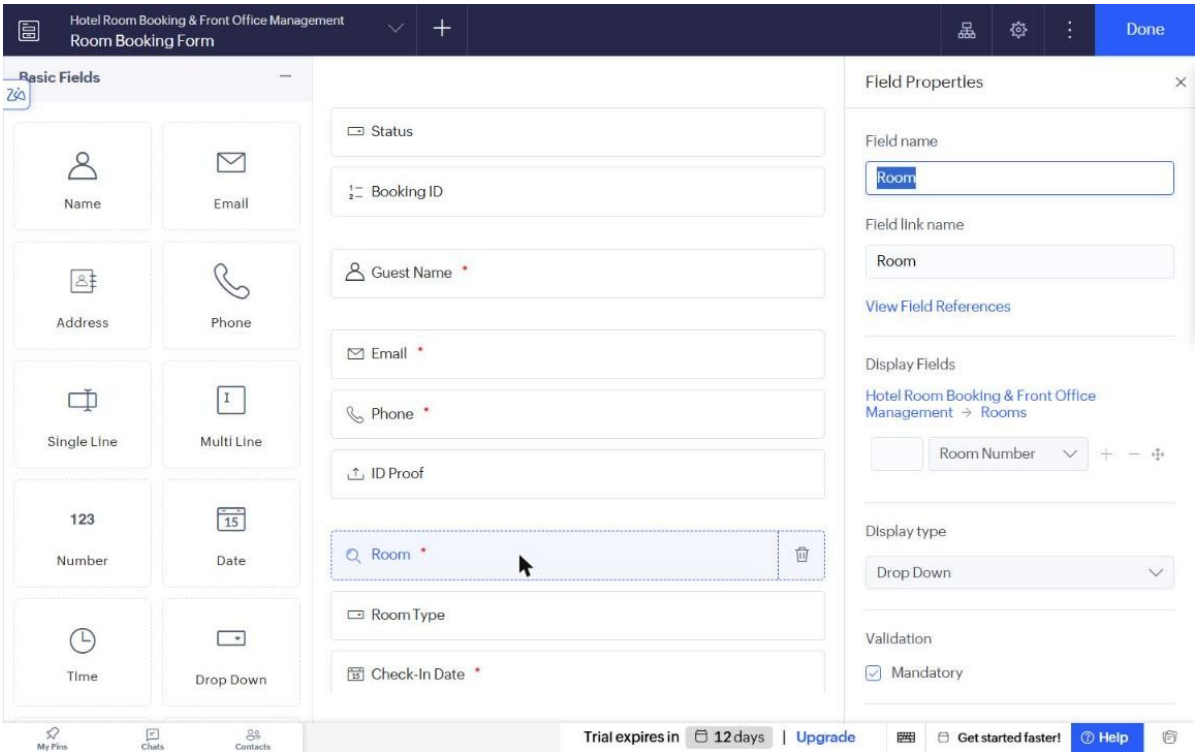


Fig. 4b.1: Room lookup field linked to the Rooms form

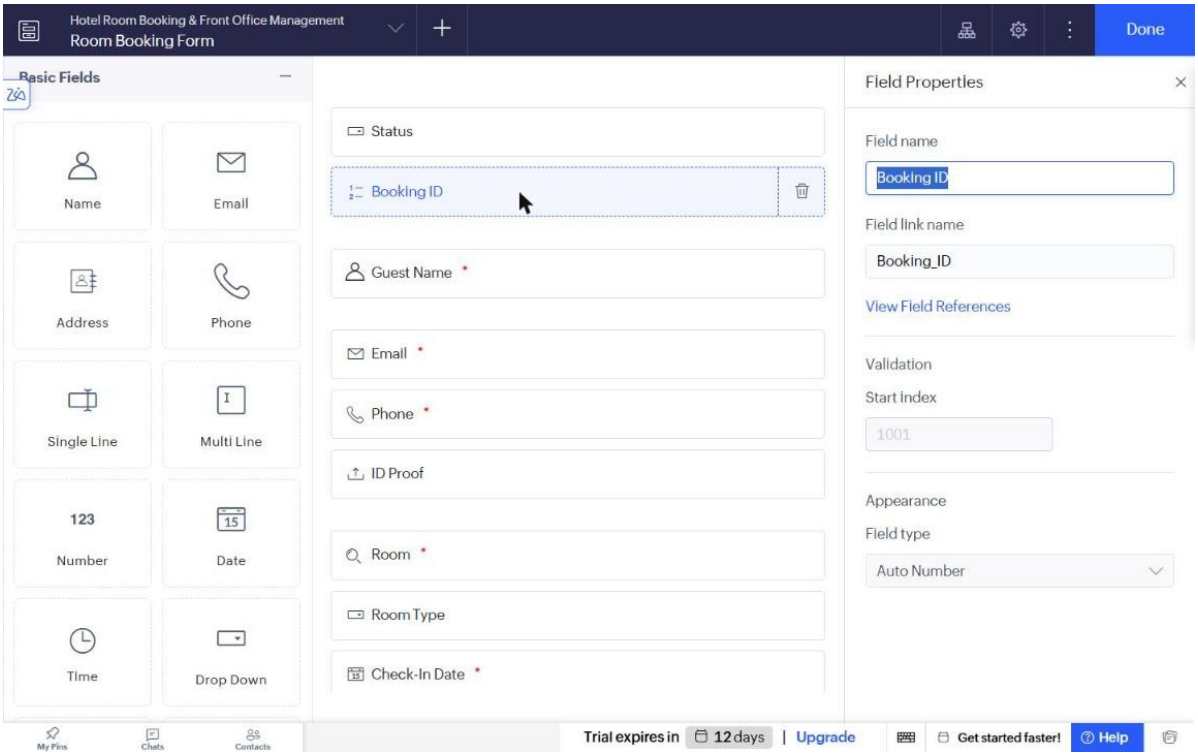


Fig. 4b.2: Booking ID - Auto Number field starting at 1001

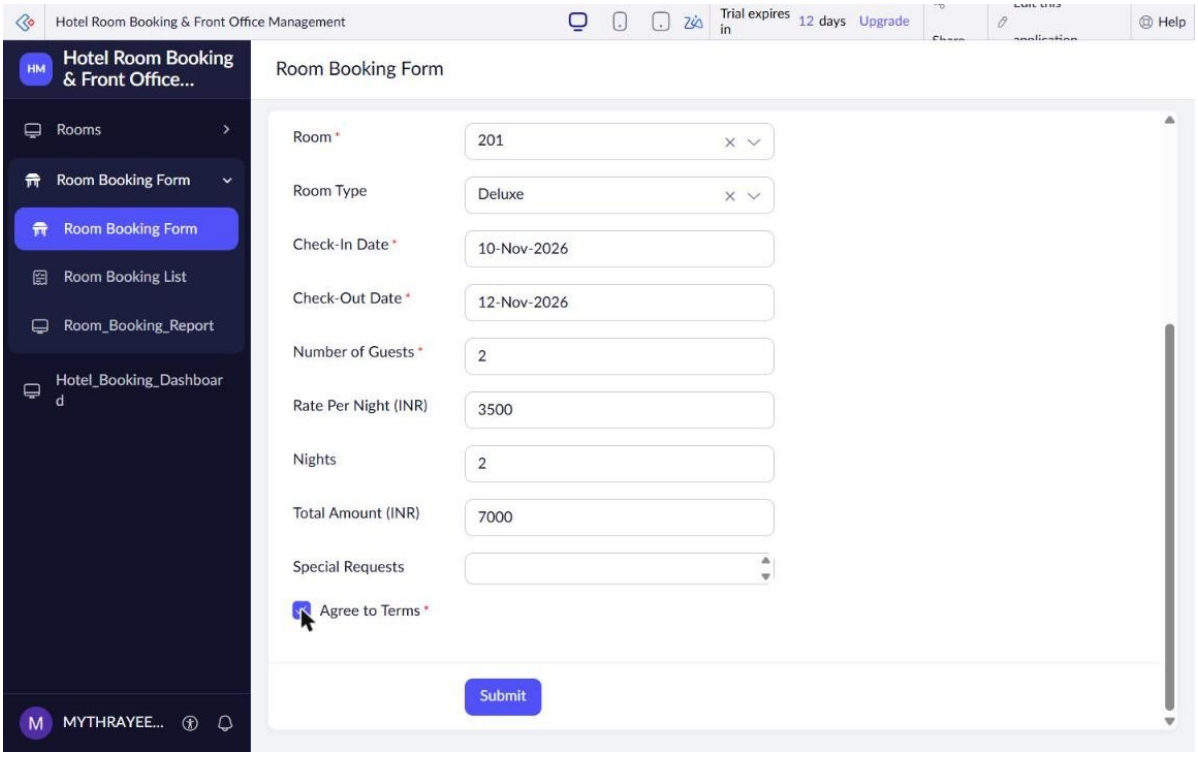


Fig. 4b.3: Room 201 selected - Room Type, Rate, Nights and Total filled automatically

Observation:

Room	Room Type	Rate (INR)	Nights	Total (INR)
201	Deluxe	3500	2 (10-12 Nov)	7000

Result:

The Room Booking Form was linked to the Rooms form through a Lookup field, Booking ID was generated automatically, and room details and the bill were calculated automatically, reducing duplicate data entry.